

Surya K Ratheesh

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OBJECTIVE

Dedicated Artificial Intelligence and Machine Learning final-year student passionate about deep learning, Explainable AI (XAI), and data science, seeking an entry-level role in machine learning engineering or data analysis..

EDUCATION

Mangalam College of Engineering <i>BTech in Artificial Intelligence And Machine Learning</i>	Kottayam, Kerala 2022 – Present
Army Public School [CBSE] <i>12th Percentage: 82.2%</i>	Kannur, Kerala 2020 - 21
Army Public School [CBSE] <i>10th Percentage: 82%</i>	Kannur, Kerala 2018 - 19

PROJECTS

Kidney Tumor Detection Web App | *Python, Streamlit, Keras*

Jan 2026 – Present

- Architected an end-to-end pipeline using OpenCV and CLAHE to automate image preprocessing and boost feature extraction accuracy, deploying the system via an intuitive Streamlit frontend for real-time visualization.
- Integrated Explainable AI (XAI) frameworks to generate diagnostic heatmaps of affected areas, transforming complex model predictions into transparent, clinician-trusted insights.

Construction Visualizer | *HTML, CSS, Javascript*

Apr 2025 – May 2025

- Engineered a web-based architectural visualizer using CSS absolute positioning and Z-index management to orchestrate multi-layered rendering of 2D assets onto building photos.
- Developed an interactive coordinate-based placement system and a responsive sidebar catalog to streamline the comparison of design configurations with instant visual feedback.

Career Guidance Chatbot | *Python, Streamlit, Pandas*

Jan 2025 - Apr 2025

- Engineered a robust, logic-based recommendation engine utilizing Pandas to dynamically filter student profiles and map academic trajectories against a comprehensive dataset of 100+ career paths.
- Deployed a conversational Streamlit interface that simplifies complex decision-making into an intuitive user journey, providing tiered guidance on entrance exams and target institutions.

Home2Work - Vehicle Hiring Platform | *TypeScript, Next.js, CSS*

Feb 2026

- Developed a responsive travel booking platform using Next.js 14, TypeScript, and Tailwind CSS, implementing dark mode, multi-page routing, and dynamic components for services, testimonials, and contact forms.
- Built a client-side contact form integrated with FormSubmit.co API, enabling email submissions to business inboxes without backend authentication, and designed a responsive UI with mobile navigation and theme switching.

SKILLS & INTERESTS

Languages: Python, C, MySQL, HTML/CSS
Developer Tools:VS Code, Visual Studio, Cursor
Soft Skills: Leadership, Teamwork, Communication Skill

LEADERSHIP EXPERIENCE

Under Officer in National Cadet Corps [NCC] 2022 - 2024

CERTIFICATIONS

- **District Winner, Young Innovators Programme 7.0:** Honored by K-DISC for excellence in the Assistive Technologies and Wearables category.
- **Python Specialization:** Successfully completed a comprehensive Python programming course via Kaggle to strengthen core data science foundations.
- **Self-Driving Electric Vehicles:** Completed an intensive technical workshop organized by Techmaghi in association with Kshitij, IIT Kharagpur, focusing on autonomous EV technologies.
- **Generative AI (Gen-AI):** Participated in a specialized IEEE webinar exploring the emerging landscape of Generative AI and its industry applications.
- **Principal's League Awardee:** Recognized for academic excellence by securing an 8.41 CGPA in B.Tech Artificial Intelligence & Machine Learning (2022-23).